

Erdős Problem #783

EP-783

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Fix some constant $C > 0$ and let n be large. Let $A \subseteq \{2, \dots, n\}$ be such that $(a, b) = 1$ for all $a, b \in A$ and $\sum_{n \in A} \frac{1}{n} \leq C$.

What choice of such an A minimises the number of integers $m \leq n$ not divisible by any $a \in A$? Is this minimised by letting $n \geq q_1 > q_2 > \dots$ be the consecutive primes in decreasing order and choosing $A = \{q_1, \dots, q_k\}$ where k is maximal such that $\sum_{i=1}^k \frac{1}{q_i} \leq C$?

"difficulty": "L1"

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>